

electron/compare-runner.cjs

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Overview

A new file `electron/compare-runner.cjs` (added lines 1-216) introduces a module that wraps a Rust-based compare engine. It discovers the binary, sanitises the environment, and offers both synchronous and asynchronous execution paths.

Key changes

- **Imports** (added lines 3-5): `const fs = require("node:fs"); const path = require("node:path"); const { spawn, spawnSync } = require("node:child_process");`
- `cliExecutableName()` (lines 7-8): returns `"novadiff-cli"` or `"novadiff-cli.exe"` based on `process.platform`.
- **resolveRustCli(appRoot, isPackaged)** (lines 15-39): searches for the binary in `process.env.NOVADIFF_CLI`, packaged resources, and dev build directories, returning the path or `null`.
- **cleanEnv()** (lines 42-46): clones `process.env` and removes `ELECTRON_OVERRIDE_DIST_PATH` and `ELECTRON_RUN_AS_NODE`.
- **runCompareEngine()** (lines 54-87): spawns the binary synchronously with `spawnSync`, checks exit status, parses JSON output, and throws detailed errors.
- **parseEngineStdout(stdout)** (lines 89-97): safely parses stdout into JSON, throwing on failure.
- **runCompareEngineAsync()** (lines 104-209): spawns the binary asynchronously, streams stdout/stderr, emits heartbeat progress events every 450 ms, and resolves with parsed JSON or rejects on error.
- **Exports** (lines 212-216): `runCompareEngine`, `runCompareEngineAsync`, `resolveRustCli`.

Impact

- **Robustness:** Centralised binary resolution and error handling reduce the chance of silent failures.
- **Responsiveness:** The async wrapper keeps Electron IPC responsive while the engine runs.
- **Cross-platform:** Handles Windows (`.exe`) and POSIX binaries, and supports both packaged and development builds.

Risks & follow-ups

- **Binary discovery:** Verify `resolveRustCli` locates the correct binary in all target environments (dev, packaged, CI).
- **Environment sanitisation:** Ensure `cleanEnv` does not strip variables required by the Rust binary.

- **Async progress:** Test that `onProgress` callbacks receive accurate `bytesReceived` and elapsed time, especially for large payloads.
- **Error paths:** Confirm that all thrown errors surface correctly in the UI; stack traces are informative.